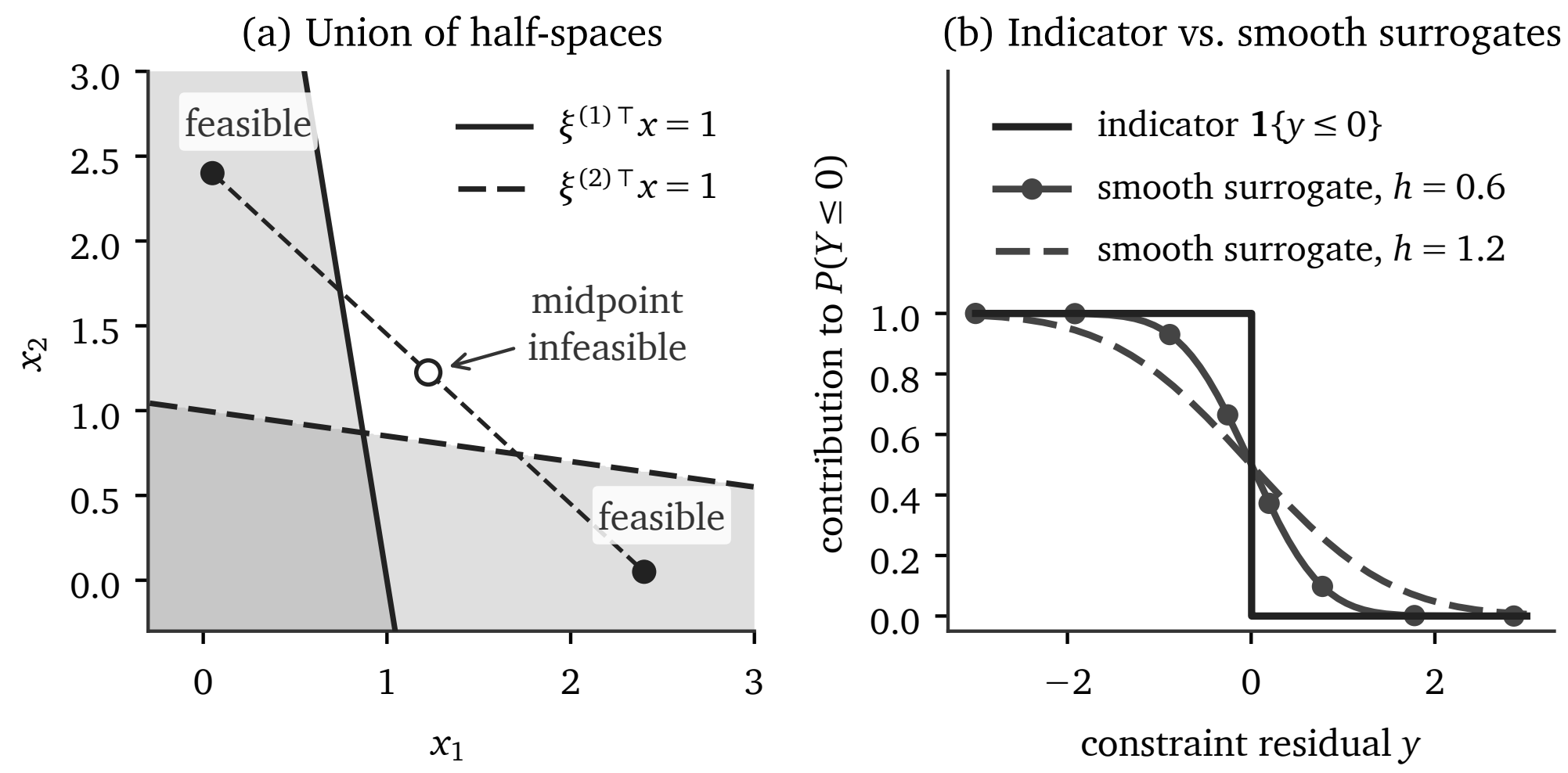
(a) Union of half-spaces
feasible
$\xi^{(1)\top} x = 1$
$\xi^{(2)\top} x = 1$
midpoint
infeasible
feasible
x_2
x_1
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
1
2
3
(b) Indicator vs. smooth surrogates
indicator $\mathbf{1}\{y \le 0\}$
smooth surrogate, $h = 0.6$
smooth surrogate, $h = 1.2$
contribution to $P(Y \le 0)$
1.0
0.8
0.6
0.4
0.2
0.0
−2
0
2
constraint residual y